

Chimera Collapse Ages: Topology-Dependent Finite-Size Scaling in Mean-Field and Ring Oscillator Systems

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Abstract

We report a long-duration empirical study of chimera collapse in two canonical settings: the two-population Sakaguchi–Kuramoto system with strictly identical natural frequencies, on the oscillator engine of a deployed additive synthesizer at small population sizes ($N = 4\text{--}64$ per population), and the nonlocally coupled ring near its chimera existence boundary. In the mean-field system, the operational collapse detector — and the first-passage criterion standard for chimera lifetimes — measure the wrong observable: up to 98% of sustained near-synchrony crossings recover as the chimera re-forms, and 69–97% of detector firings are self-healing *grazes*. Under an absorption-grade label, true lifetimes at the operating corner ($A = 0.5$, $\beta = 0.05$) double, become N -independent ($\tau_{\text{abs}} \approx 139\text{ s}$ over a $16\times$ range of N), and age (Weibull shape $2.3\text{--}3.0$, surviving stratification on the collective initial condition). Deaths are breath-synchronized and ratchet: per-cycle envelope maxima grow monotonically to capture. The reduced two-population model of Abrams *et al.*, validated against its published bifurcations with no fitted parameters, explains the inversion — the operating corner lies beyond the homoclinic, so the deployed “chimera” is a transient spiraling out along a destroyed limit cycle’s ghost, while the nominally unreliable corner ($A = 0.2$) holds the only true attractor — yet captures $3.2\times$ faster than the finite- N system dies; a direct test rules out additive collective noise, leaving that prolongation mechanism open. On the ring ($\beta = 0.110\text{--}0.130$, $N = 32\text{--}256$, **7,500** pre-registered runs), collapse is again an aging process: the Weibull shape exceeds one in every (β, N) cell and rises with N before saturating to a finite $k_\infty \approx 1.35\text{--}1.65 > 1$ (bootstrap intervals excluding one) that grows toward criticality. The dying object is a genuine chimera in **97–98%** of runs and the result is robust to the death criterion — but median ring lifetimes *decrease* with N on the near-boundary side, opposite to the mean-field

plateau. The aging of chimera collapse is topology-independent; its finite-size scaling is topology-dependent.

Keywords: chimera states, coupled phase oscillators, Kuramoto–Sakaguchi model, nonlocal coupling, finite-size scaling, transient dynamics, Weibull aging hazard

Chimera states, in which a population of identical oscillators spontaneously splits into one synchronized group and one disordered group, are usually studied as a question in pure mathematics. Here the same phenomenon appears inside a working music synthesizer, where two coupled oscillator banks shape the sound and the software periodically judges the pattern “dead” and restarts it. Asking why it dies so often reveals that the restart trigger was watching the wrong quantity: most apparent deaths are momentary lapses that heal on their own. Once the genuine deaths are isolated, they prove regular, rhythmic, and no more frequent in large populations than small ones — and a simple reduced model explains where and why they occur, while underpredicting how long the pattern survives, a discrepancy the paper leaves as an open problem. A second, classical test bed — a ring of oscillators coupled to their near neighbors — shows the same aging of collapse risk but the opposite dependence on population size, separating what is universal about chimera death from what belongs to the network’s shape.

1 Introduction

Chimera states — the coexistence of synchronized and desynchronized populations in systems of identical coupled oscillators — have been a central object of nonlinear dynamics since their discovery by Kuramoto and Battogtokh [1] and naming by Abrams and Strogatz [2]. The two-population model of Abrams, Mirollo, Strogatz, and Wiley [3] made the phenomenon solvable: two identical populations with stronger intra- than inter-population coupling support stable chimeras, breathing chimeras born in a Hopf bifurcation, and a homoclinic boundary beyond which no chimera attractor survives. At finite size the picture is enriched by transience: on nonlocally coupled rings, chimeras are chaotic transients whose lifetimes grow with oscillator count [4], and the finite- N fate of chimeras has remained a recurring theme of the literature since [5, 6], including for two small populations of the kind studied here [7].

This paper studies chimera collapse in an unusual arena: the oscillator engine of a deployed real-time additive synthesizer, in which a two-population Sakaguchi–Kuramoto bank with *strictly identical* natural frequencies drives the partial amplitudes of a musical voice, and a supervisor process re-seeds the chimera whenever an engineering detector declares it dead. Coupled-oscillator networks have an established history as sound-synthesis drivers [8, 9]; here the synthesis application matters to the dynamics only in that it fixes the regime of interest — small populations ($N = 4$ – 64 per side), long unattended runtimes, and a specific operating corner chosen, as it turns out, for the wrong reason. The deployment prompts a question the theory literature rarely needs to ask operationally: *what exactly is the death of a chimera, and what does a detector of it actually detect?*

The deployment fixes only the mean-field half of a larger question: which features of chimera collapse are universal, and which are properties of the coupling topology? The two-population mean field and the nonlocally coupled ring are the two canonical chimera substrates, and they make opposite finite-size predictions in their classic regimes — ring chimera lifetimes grow with oscillator count [4], while Sec. 4 shows the mean-field lifetimes are flat in N . We therefore close the paper by carrying the same survival methodology to the ring near its existence boundary (Sec. 7), where lifetimes are finite at feasible compute, asking whether the aging structure of collapse — the rising hazard — transfers across topology even where the lifetime scaling does not.

We answer with a chain of measurements, each gating the next, on bit-reproducible integrators (2,100 baseline mean-field runs plus targeted re-campaigns, and 7,500 pre-registered ring runs; every figure regenerable from committed configurations). The contributions are:

1. **Graze versus absorption.** The operational collapse criterion — sustained near-synchrony of the incoherent population — overwhelmingly fires on events the chimera survives: up to 98% of criterion-satisfying crossings recover, and 69–97% of the deployed detector’s firings are such grazes. We introduce a two-timescale, absorption-grade labeling and show first-passage “lifetimes” are roughly half the true absorption times (Sec. 3).
2. **An inverted stability picture.** Under absorption labeling, lifetimes at the operating corner are N -independent (≈ 139 s, $N = 4$ –64, every seed absorbing), while at the nominally unreliable high-morph corner the majority of seeds *never* absorb: the operating corner has no chimera attractor at all, and the discarded corner holds the only persistent chimera in the system (Sec. 4).
3. **The death mechanism.** Collapse is a breath-synchronized ratchet: collapse attempts and true deaths lock to a phase of the breathing cycle, successive per-cycle envelope maxima grow monotonically (spiraling out exponentially toward the synchronization boundary), and hazard rises with cycle index. Transverse escape from the Poisson submanifold is ruled out by a validated observable and a null interventional test; lifetime is instead organized by the collective initial condition, dominantly the inter-population phase offset (Sec. 5).
4. **Closure by the reduced model — and a clean open problem.** The reduced equations of Ref. [3], validated against that paper’s own published results and applied with a literal 1:1 timescale and zero fitted parameters, place the operating corner beyond the homoclinic ($A_{hc}(\beta=0.05) = 0.4096$), reproduce the breath period and per-cycle spiral rates, identify the never-absorbers’ level with the stable chimera’s r^* , and predict individual run lifetimes beyond any static feature of the seed. They also fail in one instructive way: mean-field capture is $3.2\times$ faster than finite- N death, and the flat $\tau_{\text{abs}}(N)$ is absent from the N -free collective flow — finite-size fluctuations *extend* the transient (Sec. 6, Sec. 8).
5. **Topology comparison: aging transfers, scaling does not.** On the nonlocally coupled ring near its existence boundary, collapse is also an aging process: the censored-Weibull shape exceeds one, with confidence intervals excluding one, in all 15 pre-registered (β, N) cells, and rises with N before saturating to a finite $k_\infty \approx 1.35$ – $1.65 > 1$ that grows toward criticality (Appendix C). The dying object

is validated as a genuine chimera in 97–98% of runs and the hazard shape is robust to the death criterion. Median ring lifetime, however, *decreases* with N in this regime — opposite to both the deep-stable ring growth [4] and the mean-field plateau: the aging of chimera collapse is topology-independent, while its finite-size scaling is topology-dependent (Sec. 7).

2 Systems

2.1 The two-population mean-field system

The voice integrates two populations $\sigma \in \{1, 2\}$ of N phase oscillators each,

$$\dot{\theta}_i^\sigma = \omega + \sum_{\sigma'} \frac{K_{\sigma\sigma'}}{N} \sum_{j=1}^N \sin(\theta_j^{\sigma'} - \theta_i^\sigma - \alpha), \quad (1)$$

with intra-population coupling $K_{\sigma\sigma} = \mu = (1 + A)/2$, inter-population coupling $K_{\sigma\sigma'} = \nu = (1 - A)/2$ ($\mu + \nu = 1$, $A = \mu - \nu$), and phase lag $\beta = \pi/2 - \alpha$ [10, 11], matching the conventions of Ref. [3]. A code audit confirms the natural frequencies are identical *exactly* ($\delta\omega = 0$: no per-oscillator frequency term, detuning explicitly zeroed on this path), so the dynamics is deterministic given the initial condition and collapse cannot be attributed to frequency disorder; we work in the $\omega = 0$ rotating frame. Integration is fourth-order Runge–Kutta with $dt = 0.05$ at a documented one-model-unit-per-second clock, so all times below are seconds with no conversion; halving and quartering dt leaves lifetime distributions statistically indistinguishable (two-sample KS $p = 1.0$). All runs are bit-reproducible from logged (seed, parameters, dt); every re-analysis below begins by reproducing the upstream labels exactly.

In the application, the incoherent population’s per-oscillator coherences modulate the gains of an additive partial bank, $g'_i = g_i [1 + da(c_i - \frac{1}{2})]$ with $c_i = \frac{1}{2}(1 + \cos(\theta_i - \psi))$, turning the chimera’s collective dynamics into a slowly self-morphing spectrum; the synthesis layer plays no further role in the dynamics studied here. Runs are initialized with the canonical chimera seed (one population tightly clustered, the other spread), since spontaneous chimera formation from random phases is negligible in this regime. The deployed supervisor declares the state alive while $\max_\sigma R_\sigma > 0.9$ and $\min_\sigma R_\sigma < 0.85$, fires after 2 s of sustained violation, and re-seeds the incoherent population. The baseline campaign comprises 200 seeds per population size at the operating corner ($A = 0.5$) and 100 at the high-morph corner ($A = 0.2$), $N \in \{4, 8, 16, 24, 32, 48, 64\}$, $\beta = 0.05$ throughout, integrated unsupervised to $t_{\max} = 2000$ s.

2.2 The nonlocally coupled ring

The second system is the canonical substrate of chimera-death studies, the nonlocally coupled ring [1, 4]: N phase oscillators with top-hat coupling to their P nearest

neighbors on each side,

$$\dot{\theta}_k = -\frac{1}{2P+1} \sum_{|j-k| \leq P} \sin(\theta_k - \theta_j + \alpha), \quad P = \text{round}(rN), \quad (2)$$

with coupling radius $r = 0.15$, phase lag $\alpha = \pi/2 - \beta$, identical frequencies in the rotating frame, and the same fixed-step RK4 integration scheme (validated against an independent reference implementation). The natural order parameter is local: $\rho_k = |(2P+1)^{-1} \sum_{|j-k| \leq P} e^{i\theta_j}|$, the modulus of the windowed mean field, whose spatial profile separates a chimera’s coherent arc ($\rho_k \approx 1$) from its incoherent arc (low ρ_k). Death of the ring chimera is the loss of this spatial structure: the event criterion is the spatial standard deviation $\rho_{\text{std}}(t)$ of the ρ_k field falling below $\varepsilon_{\text{std}} = 0.04$ and staying below for 50 time units, with right-censoring at $T_{\text{max}} = 12,000$; Sec. 7 validates this criterion against alternatives and against the identity of the dying object. The ring campaign — design, event definition, primary endpoint (the censored-Weibull shape k with profile-likelihood intervals), and decision rule all pre-registered before any production run — comprises $\beta \in \{0.110, 0.115, 0.120, 0.125, 0.130\}$ approaching the existence boundary $\beta_c \approx 0.13$ – 0.14 , at $N \in \{32, 64, 128\}$ with 300 runs per cell (4,500 runs), later extended at all five β to $N = 192$ and 256 (3,000 further runs) for the thermodynamic-limit extrapolation of Appendix C.

3 Graze versus absorption

What counts as the death of a chimera? In a deployed system the answer is operational: the supervisor’s aliveness condition fails continuously for 2 s, and our initial lifetime campaign used the closely related first-passage criterion that $\min(R_1, R_2)$ exceed a threshold $\theta = 0.85$ sustained for $W = 5$ s. Both definitions encode the same assumption: that a sustained excursion of the incoherent population to near-synchrony *is* the death of the chimera. This section shows that assumption is wrong, and replaces it.

3.1 Threshold crossings overwhelmingly recover

Extending the recorded trajectories past their first criterion-satisfying crossing, rather than terminating there, reveals that the crossing is usually not absorbing. At the operating corner and across $N = 8$ – 64 , up to 98% of first crossings — excursions exceeding θ for longer than the full 5 s sustain window — are followed by recovery: the incoherent population desynchronizes again, with post-crossing minima of $\min(R_1, R_2)$ clustering at 0.2–0.45, a fully re-formed chimera. About two-thirds of these recovered trajectories are later absorbed for good; the remainder churn through further graze–recover cycles. The first sustained crossing is therefore a *graze* of the synchronization boundary, not a death, and first-passage lifetimes computed from it measure time-to-first-long-graze.

Figure 1 shows a representative trajectory ($N = 16$, $A = 0.5$; seed pinned in the released configuration). The run grazes twice — each time exceeding θ for more than 5 s before the chimera re-forms — and is absorbed on its third approach at $t_{\text{abs}} = 221.9$ s, having first “collapsed” by the engineering criterion at $t_{\text{graze}} = 83.2$ s.

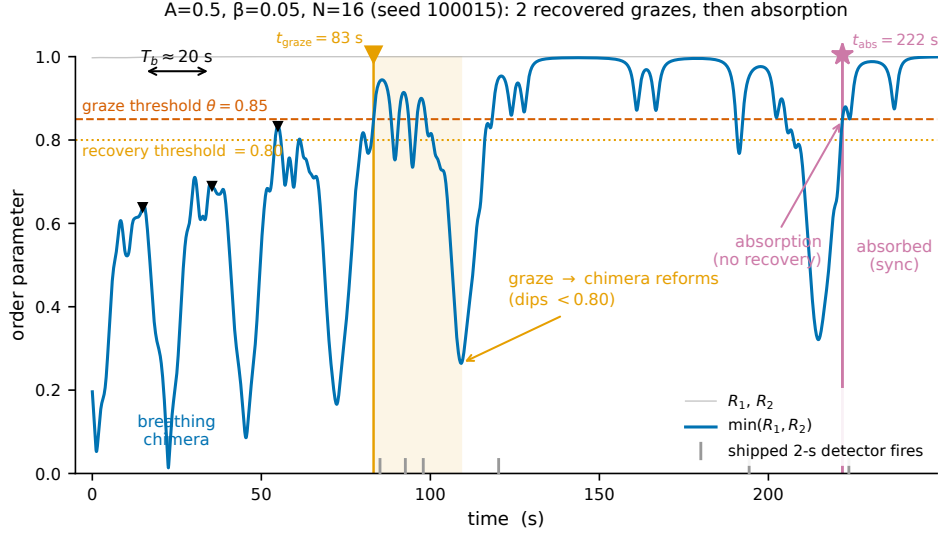


Fig. 1 A graze is not a death. $\min(R_1, R_2)$ for a single run ($N = 16$, $A = 0.5$, $\beta = 0.05$; breath period $T_b = 20.5$ s). The trajectory crosses $\theta = 0.85$ (upper line) in sustained excursions twice, recovering below the 0.80 recovery threshold (lower line) each time as the chimera re-forms, before the absorbing crossing at $t_{\text{abs}} = 221.9$ s. The engineering first-passage label $t_{\text{graze}} = 83.2$ s and the six firings of the deployed 2-s detector are marked.

The shipped supervisor’s detector fires *six times* on this trajectory before the single event that is irreversible: six firings, two reforms, one death.

3.2 An absorption-grade label

We therefore assign every run two timescales from a single trace. The first, t_{graze} , is the unmodified first-passage label, retained for comparability. The second, t_{abs} , is the time of the first crossing *not* followed by recovery — $\min(R_1, R_2) < 0.80$ sustained at least 5 s — within a verification horizon $T_v = 120$ s; runs still alternating between grazes and recoveries at the integration horizon are right-censored.

The absorption label is robust to its own knobs where it matters. Sweeping $T_v \in \{60, 120, 240\}$ s against recovery thresholds $\{0.75, 0.80\}$ moves $\hat{t}_{\text{abs}}$ by 22–27% at $A = 0.5$, monotonically in T_v and negligibly in the recovery threshold — comparable to or better than the $\sim 52\%$ (θ, W) -sensitivity of the graze criterion itself. At $A = 0.2$ the apparent sensitivity is dominated by censoring rather than the labeling parameters, for a reason Sec. 4 shows is dynamical: most $A = 0.2$ trajectories never absorb at all.

The quantitative consequence at the operating corner is a factor of two: $\hat{t}_{\text{graze}} \approx 63\text{--}67$ s against $\hat{t}_{\text{abs}} \approx 139$ s (Fig. 2). First-passage chimera lifetimes computed with engineering-style criteria systematically measure the wrong, and roughly halved, quantity.

Table 1 Supervisor over-trigger: fraction of the deployed 2-s detector’s firings, replayed over unsupervised traces, occurring on self-recovering grazes.

A	N	firings/run	over-trigger	median wasted (s)
0.5	8	7.2	79%	4.4
0.5	16	5.2	76%	2.4
0.5	32	4.1	73%	3.9
0.5	64	3.3	69%	5.3
0.2	8	12.3	95%	2.0
0.2	16	9.8	97%	1.8
0.2	32	7.9	96%	1.7
0.2	64	4.7	95%	1.8

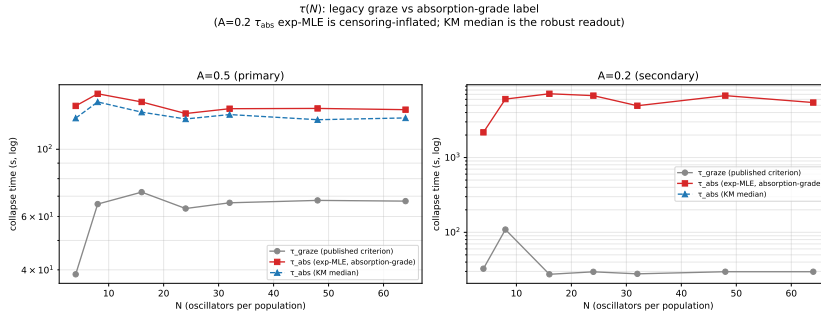


Fig. 2 Graze versus absorption lifetimes. Exponential-MLE $\hat{\tau}_{\text{graze}}$ and $\hat{\tau}_{\text{abs}}$ versus N at both coupling disparities. At $A = 0.5$ absorption lifetimes are $\approx 2.2\times$ the first-passage values and flat in N ; at $A = 0.2$ no absorption lifetime is defined for the majority of seeds (Sec. 4).

3.3 The deployed detector is a graze detector

Replaying the shipped detector over the same unsupervised trajectories and classifying each firing by whether the underlying event self-recovers gives Table 1: 69–79% of firings at $A = 0.5$, and 95–97% at $A = 0.2$, occur on grazes the chimera survives on its own. The deployed system has been re-seeding a state that was mostly not dead — harmless in practice, since re-seeding lands near where the trajectory already was, but diagnostic of the conceptual gap this section closes: engineering collapse detection and dynamical absorption are different observables. All statistics below are computed on t_{abs} unless noted.

4 Absorption statistics at finite N

4.1 A flat plateau at the operating corner

At $A = 0.5$, every one of the 1,400 seeds absorbs within the horizon (0% censored at every N), and the absorption lifetime is N -independent: $\hat{\tau}_{\text{abs}} = 139, 152, 143, 131, 136, 136, 135$ s for $N = 4\text{--}64$, a ratio $\tau(64)/\tau(4) = 0.97$ over a sixteen-fold range of population size, with fitted exponential rate in N indistinguishable from zero. The weak rise from $N = 4$ to 16 visible in the graze times ($39 \rightarrow 72$ s)

does not survive absorption labeling — it was a graze artifact. The published-style picture of a lifetime that grows with oscillator count, familiar from ring-topology chimeras [4], is absent here in both label and shape: adding partials does not buy persistence at this corner, at any tested N .

4.2 The inversion: the high-morph corner does not absorb

The corner the application had classified as unreliable behaves in the opposite way. At $A = 0.2$, between 51% and 78% of seeds (rising with N) *never absorb*: extending the horizon eightfold to $t_{\max} = 16,000$ s leaves the censored fraction essentially unchanged, so the censoring is irreducible — a dynamical finding, not a horizon artifact. The minority that does absorb does so promptly and increasingly on the first approach: the zero-graze fraction among absorbers rises from 0.08 at $N = 4$ to 0.96 at $N = 64$. Capture at $A = 0.2$ is bimodal — absorb almost immediately, or settle into a persistent, intermittently grazing chimera [12] with no defined absorption time. The stability assumption of the deployment is therefore inverted: the “reliable” corner always dies, and the “risky” corner contains the only persistent, non-absorbing chimeras in the system. Section 6 shows both facts are exactly where the bifurcation diagram says they should be.

4.3 Aging: hazard rises with every breath

Absorption at $A = 0.5$ is not memoryless. Censored-Weibull fits give shape parameters k_{abs} of 2.27, 2.22, 2.49, 2.67, 2.44, 2.72, and 3.03 for the seven sizes $N = 4$ –64, all confidence intervals well above 1 (Fig. 3) — stronger aging than in the graze times ($k_{\text{graze}} = 0.82$ –2.19), so the survival shoulder is not a labeling artifact.

Nor is the aging an artifact of pooling heterogeneous deterministic transients across the seed ensemble. Because the dynamics is deterministic, a fitted $k > 1$ could in principle encode only the *spread of fixed transient times* across initial conditions rather than a hazard that rises along each trajectory. We separate the two by stratifying the seed ensemble at each N into four equal-count bins of a collective initial-condition coordinate — the inter-population phase gap $|\Delta\phi_0|$, the initial incoherent-population level $\bar{R}_{\text{incoh},0}$, or, decisively, the reduced model’s own predicted lifetime for that seed (Sec. 6), which collapses the entire collective initial condition onto its lifetime consequence — and refitting the censored Weibull within each stratum. The aging survives in every case: $k_{\text{abs}} > 1$ with 95% profile-likelihood intervals excluding one in all 84 size×stratum cells, with $k_{\text{abs}} \in [2.00, 5.00]$ under the predicted-lifetime binning (likelihood-ratio $p \leq 10^{-7}$ against the exponential throughout), and the within-stratum lifetime spread remains substantial ($\text{CV} = 0.22$ –0.56) while tracking the fitted shape through the Weibull CV– k relation at correlation 0.996 — a genuine aging-shaped spread, not a degenerate spike. Censoring at $A = 0.5$ is exactly zero (all 1,400 runs absorb before $t_{\max}$), so these are effectively uncensored maximum-likelihood fits. The aging is trajectory-level — a hazard rising along each run, as the ratchet of Sec. 5 shows directly — not initial-condition heterogeneity.

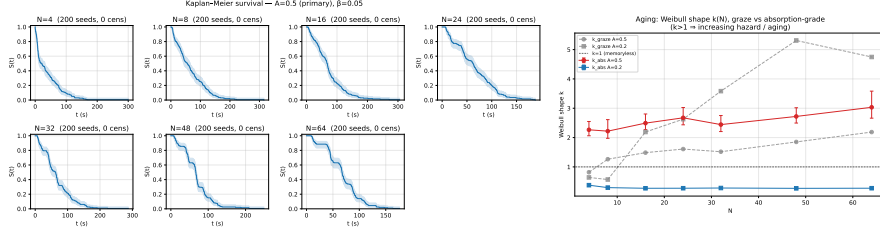


Fig. 3 Absorption survival and aging. Kaplan–Meier survival curves by N (left, $A = 0.5$) develop a shoulder with increasing N ; censored-Weibull shape $k_{\text{abs}}(N)$ (right) rises from 2.3 to 3.0 with all confidence intervals above the memoryless value $k = 1$.

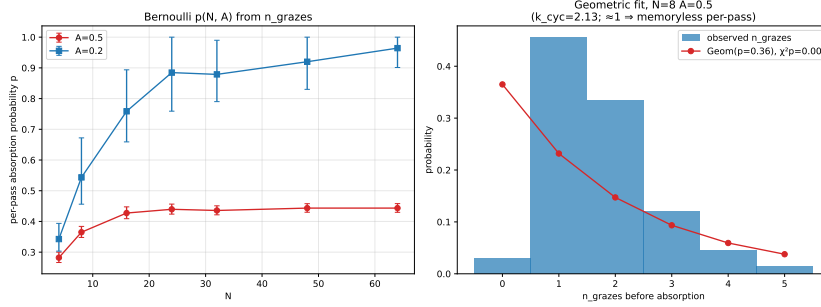


Fig. 4 Per-pass absorption statistics at $A = 0.5$. Mean per-pass absorption probability $\hat{p}(N)$ rises then plateaus, mirroring $\tau_{\text{abs}}(N)$; the geometric (constant- p) model is rejected at every point — hazard rises with cycle index ($k_{\text{cyc}} > 1$).

The natural per-breath Bernoulli model — a constant absorption probability per high- R pass — is rejected at every $A = 0.5$ point ($\chi^2 p < 0.001$ against geometric cycles-to-absorption; Weibull-in-cycles shape $k_{\text{cyc}} \approx 2.1$ –2.9): the per-pass hazard *rises* with cycle index. The average per-pass absorption probability $\hat{p}$ nonetheless tracks the plateau, rising from 0.28 at $N = 4$ to 0.44 by $N = 24$ and flattening (Fig. 4), while the mean number of recovered grazes before death falls from 2.5 to 1.25. At $A = 0.2$ the strong apparent aging in the graze statistics (k_{graze} up to 5.3) is retired: on the few true absorptions $k_{\text{abs}} \approx 0.3 < 1$, a fast, front-loaded minority rather than aging. The never-absorbers hold a mean incoherent-population level $\bar{R}_{\text{incoh}} \approx 0.66$ –0.68, sitting at the stable fixed point $r^* = 0.678$ identified in Sec. 6.

5 Mechanism: the breath-synchronized ratchet

5.1 Deaths lock to the breathing phase

The chimera at $A = 0.5$ breathes: $\min(R_1, R_2)$ oscillates with period $T_b \approx 21$ –25 s, its envelope reaching ≈ 0.83 against the $\theta = 0.85$ boundary — the cycle passes within a hair of near-synchrony once per breath. Collapse *attempts* (long grazes) lock to this cycle: their phases relative to the preceding envelope peak cluster at 0.13–0.20 of the cycle, with resultant length tightening from 0.43 to 0.69 as N grows and pooled

Rayleigh $p \approx 0$. True absorptions inherit the lock: pooled over $A = 0.5$, $\bar{R} = 0.21$ with Rayleigh $p = 2.2 \times 10^{-6}$ (three of six individually testable points significant; clustering weakens at large N , where capture completes within 2–3 cycles and leaves few phases to test). Mean absorption takes roughly six breaths ($\tau_{\text{abs}}/T_b \approx 6$). Death, when it comes, comes at a particular moment of the breath (Fig. 5).

5.2 The envelope ratchets monotonically to capture

Within individual runs, successive breathing maxima M_k of $\min(R_1, R_2)$ do not fluctuate about a level — they climb. The mean per-cycle increment $\langle \Delta M \rangle$ is $+0.031, +0.052, +0.078, +0.091$ for $N = 8, 16, 32, 64$, with cluster-bootstrap confidence intervals strictly positive at every point; the fraction of runs with monotone (allowing one inversion) maxima reaches 1.00 by $N = 32$. Where enough sub-threshold approach peaks exist to fit, $\log(\theta - M_k)$ falls linearly in cycle index with median slopes -0.55 to -0.77 and a negative slope in 100% of fittable runs: the approach to the synchronization boundary is an exponential spiral-out (Fig. 6). The control rules out artifacts: the same detector and windowing applied to the $A = 0.2$ never-absorbers over a median of ~ 200 cycles gives $\langle \Delta M \rangle \approx -0.001$ and a ratchet fraction of exactly zero — stationary, with a slight *relaxation* consistent with settling onto an attractor.

5.3 What collapse is not, and what organizes it

Because the oscillators are strictly identical, a natural mechanism would be escape transverse to the Poisson (Ott–Antonsen) submanifold [13, 14] on which the reduced chimera solutions live. We tested this directly with the moment observable $D = \sum_{m=2}^4 |Z_m - Z_1^m|^2$, validated in three stages (the Poisson identity $Z_m = Z_1^m$ to machine precision on Möbius-constructed states; dynamical invariance of on-manifold initial conditions under the shipped integrator; a clustered-state contrast), and the hypothesis fails comprehensively: initial distance D_0 does not predict lifetime ($|\rho| \leq 0.15$ at all fourteen parameter points), event-aligned $\langle D \rangle$ shows no ramp preceding collapse, and a paired interventional experiment injecting controlled D_0 levels leaves lifetimes flat. Collapse is not transverse escape.

It is instead organized in the collective coordinates. Among static features of the seed, the initial inter-population phase offset dominates: $|\Delta\phi_0|$ anticorrelates with lifetime at $\rho \approx -0.43$ to -0.59 for $N \geq 8$ (median magnitude 0.52), versus the D_0 null's ≤ 0.08 , and a cross-validated regression on the $t = 0$ collective features reaches $R^2 \approx 0.2$ (quadratic up to 0.44). Larger initial phase separation between the populations means earlier death — a fact Sec. 6 converts from regression to dynamics.

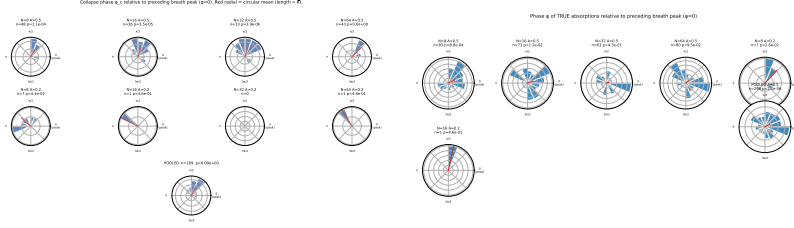


Fig. 5 Breath-phase locking. Circular histograms of collapse-attempt phases (left) and true-absorption phases (right) relative to the preceding breathing-envelope peak, $A = 0.5$. Attempts cluster at 0.13–0.20 of the cycle (pooled Rayleigh $p \approx 0$); absorptions inherit the lock (pooled $p = 2.2 \times 10^{-6}$).

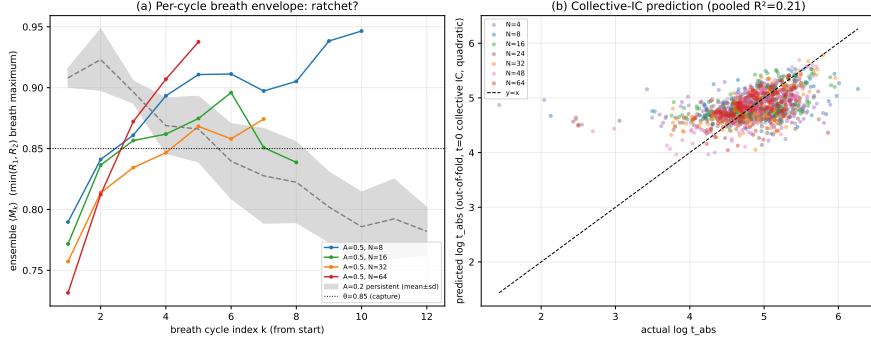


Fig. 6 The ratchet. Ensemble-averaged breathing maxima $\langle M_k \rangle$ versus cycle index k at $A = 0.5$ for several N , with the $A = 0.2$ never-absorber band (stationary, $\langle \Delta M \rangle \approx -0.001$) for contrast. Per-cycle increments are strictly positive at every N ; $\log(\theta - M_k)$ falls linearly (median slopes -0.55 to -0.77).

6 Reduced dynamics: the post-homoclinic ghost

6.1 The reduced system, validated against its source

In the $N \rightarrow \infty$ mean-field, the two-population system closes in the per-population order parameters. Writing $\rho_\sigma e^{-i\phi_\sigma}$ for population σ 's centroid, the reduced flow is [3]

$$\dot{\rho}_1 = -\frac{\rho_1^2 - 1}{2} \left[\mu \rho_1 \cos \alpha + \nu \rho_2 \cos(\phi_2 - \phi_1 - \alpha) \right], \quad (3)$$

together with the corresponding $\dot{\phi}_1$ equation and the $1 \leftrightarrow 2$ images; rotational symmetry reduces this to three variables $(\rho_1, \rho_2, \psi = \phi_1 - \phi_2)$. On the invariant manifold $\rho_1 = 1$ (synchronized population exactly locked) the flow further reduces to two variables $(r = \rho_2, \psi)$, Eq. (12) of Ref. [3], whose stationary chimeras, saddle-node curve $A_{SN}(\beta)$, and Hopf curve $A_H(\beta)$ are known in closed or series form, with the linear stability of these two-population chimeras characterized in detail by Laing [15].

Before applying any of this to our system, the implementation was gated against the source paper itself: the parametrized stationary chimeras zero the reduced right-hand side to 4×10^{-16} ; integration reproduces the published regimes at $\beta = 0.1$ (stationary at $A = 0.2$; breathing at $A = 0.28$ and 0.35 , with the longer period at the

larger A); and numerically located saddle-node and Hopf points match the published series to 5×10^{-5} . Every number below is parameter-free and, by the integrator’s documented clock, in literal seconds.

6.2 The two corners straddle the bifurcation set

At $\beta = 0.05$: $A_{SN} = 0.0948$, $A_H = 0.2703$, and the homoclinic — located by bisection on the divergence of the breathing cycle’s period — at $A_{hc} = 0.4096$ (bracket width 8.8×10^{-5}). Traced over $\beta \in [0.03, 0.20]$, the homoclinic curve descends, turns, and rises into the Takens–Bogdanov point at $(0.2239, 0.3372)$ where it meets the saddle-node and Hopf curves — the non-monotone approach is demanded by the termination and serves as an internal consistency check (Fig. 7).

The verdicts follow immediately. $A = 0.2$ lies in the stable-chimera band ($A_{SN} < 0.2 < A_H$): the reduced system has a stable stationary chimera at $r^* = 0.6781$ (integration amplitude decaying to 10^{-9}), with weak relaxation $\sigma = -0.0054 \text{ s}^{-1}$ — matching the never-absorbers, whose mean incoherent level sits at r^* and whose envelope drifts slightly *down*, and explaining the bimodal capture as a basin question: seeds either fall to synchrony on the first approach or onto the chimera attractor forever. $A = 0.5$ lies beyond the homoclinic ($0.5 > A_{hc} = 0.4096$): *no chimera attractor exists at the operating corner*. The chimera fixed point is an unstable spiral ($\sigma = +0.01243 \text{ s}^{-1}$, $\omega = 0.3277 \text{ s}^{-1}$), the breathing limit cycle that once surrounded it is destroyed, and sync is globally attracting. What the deployed voice has been operating is a transient spiraling outward along the ghost of that destroyed cycle (Fig. 8).

6.3 Quantitative match with no fitted parameters

Table 2 compares the reduced flow to the finite- N measurements. The spiral’s rotation gives a linear breath period $2\pi/\omega = 19.2 \text{ s}$ and a nonlinear period of 25.0 s for trajectories at canonical-seed amplitudes — bracketing the measured $21\text{--}25 \text{ s}$. The reduced per-cycle approach slopes, computed identically to the measurement from each run’s seed-mapped initial condition, are $-0.62, -0.66, -0.66, -0.73$ for $N = 8\text{--}64$ against measured $-0.55, -0.50, -0.59, -0.77$: the trend with N is reproduced even though the reduced flow is N -free, because the seed ensembles at different N occupy different collective initial conditions (mean $R_{\text{incoh},0}$ falls $\sim 1/\sqrt{N}$). The single substantial mismatch is the absolute capture time: the reduced median is 43 s against the measured 139 s — the mean-field flow dies $3.2\times$ faster than the finite- N system (Sec. 8). (A single trajectory started near the focus, by contrast, escapes only at $\approx 726 \text{ s}$: at $\sigma = 0.0124 \text{ s}^{-1}$ the ~ 9 e-folds from a near-focus deviation take exactly that long, while the seed ensemble starts at finite amplitude — the two timescales are the same spiral read from different starting radii.)

6.4 Individual lifetimes from a three-variable model

Finally, the reduced system predicts *individual* deaths. Mapping every $A = 0.5$ seed to its collective coordinates $(\rho_1, \rho_2, \psi)_0 = (R_{\text{sync}}, R_{\text{incoh}}, \Delta\phi)_0$ and integrating the three-variable flow to capture yields a predicted lifetime whose Spearman correlation with the measured t_{abs} is 0.445 pooled (up to 0.60 per N) — comparable to the best fitted

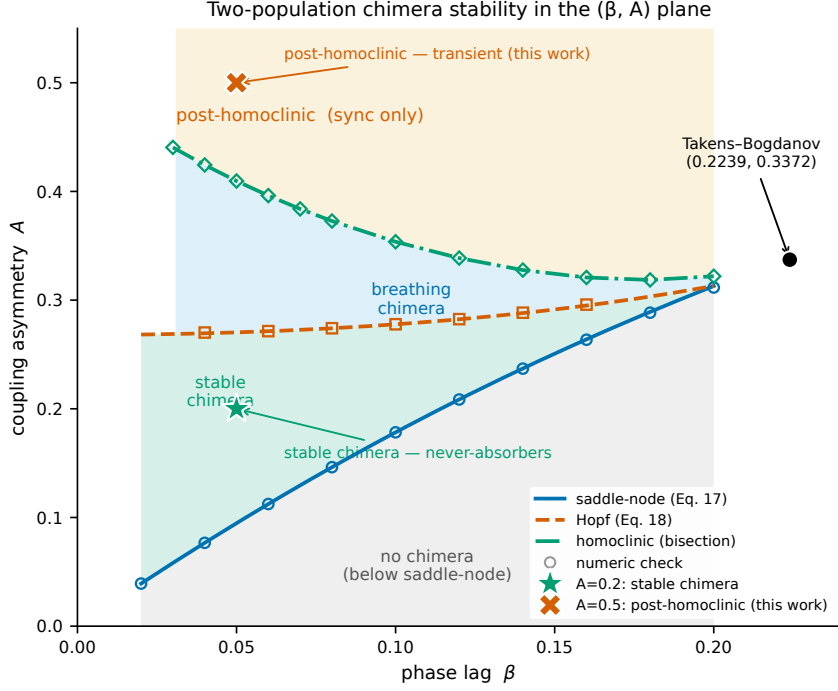


Fig. 7 Stability diagram of the two-population system in the (β, A) plane: saddle-node (Eq. 17 of Ref. [3]) and Hopf (Eq. 18) curves with numeric checks, the numerically traced homoclinic curve terminating at the Takens–Bogdanov point $(0.2239, 0.3372)$, and the two operating corners of the deployed system. The nominally stable corner ($A = 0.5$) is post-homoclinic — no chimera attractor — while the nominally unreliable corner ($A = 0.2$) sits inside the stable-chimera band.

Table 2 Reduced flow versus finite- N measurement at the operating corner ($A = 0.5$, $\beta = 0.05$). No fitted parameters; times in seconds at the integrator’s documented 1:1 clock. The ratio is reduced/measured throughput; the final row reports the $A = 0.2$ chimera level for comparison.

quantity	reduced	measured	ratio (reduced/measured)
breath period T_b	25.0 (linear 19.2)	21–25	≈ 1.1
per-cycle spiral slope (pooled)	−0.67	−0.55 to −0.77	≈ 1.0
median capture time	43	139	0.31
$A=0.2$: chimera level	$r^* = 0.678$	$\bar{R}_{\text{incoh}} = 0.66\text{--}0.68$	≈ 1.0

static feature — and, crucially, carries a *positive partial correlation* of $+0.20\text{--}0.32$ *beyond* $|\Delta\phi_0|$ *at every* N : the parameter-free dynamics predicts lifetime variance no static regression on the same inputs captures (Fig. 9). The transverse coordinate stays quiet throughout (ρ_1 deviating from 1 by ≤ 0.003 median), justifying the manifold picture a posteriori. This N -independence is not merely a mean-field artifact: for identical oscillators under global sinusoidal coupling, Watanabe–Strogatz theory [16, 17] reduces each population to a three-parameter flow at *every* N , with the remaining $N - 3$ degrees of freedom frozen as constants of motion — as the group-theoretic [18]

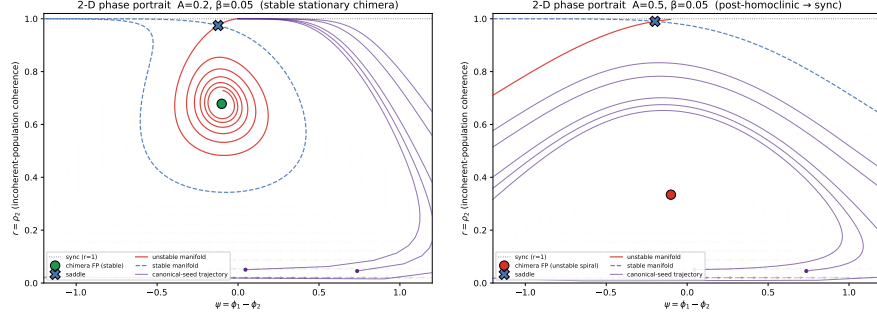


Fig. 8 Reduced-system phase portraits at $\beta = 0.05$. Left ($A = 0.2$): stable chimera spiral at $r^* = 0.678$ with the saddle and its manifolds organizing the basin. Right ($A = 0.5$): the chimera fixed point is an unstable spiral ($\sigma = +0.0124 \text{ s}^{-1}$); canonical-seed trajectories spiral out to synchronization.

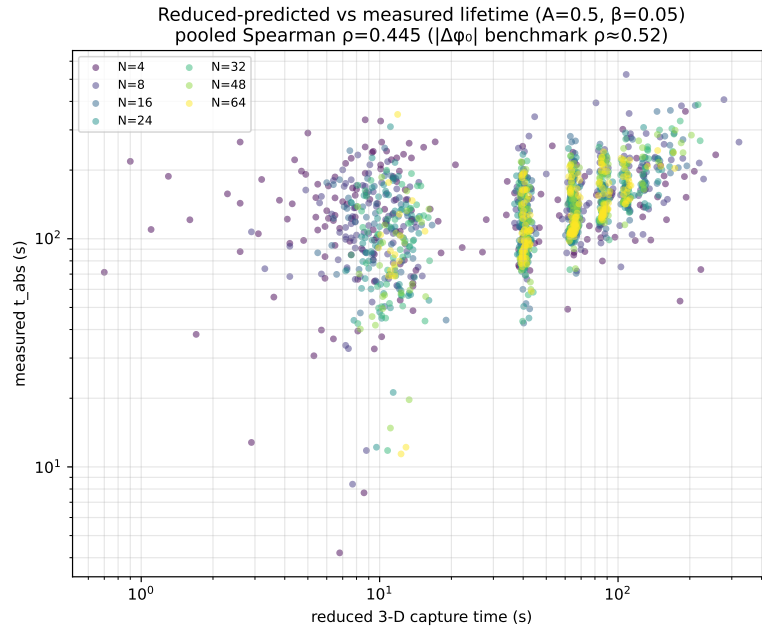


Fig. 9 Run-by-run lifetime prediction. Reduced-model capture time from each seed's collective initial condition versus measured t_{abs} , colored by N ($A = 0.5$; pooled Spearman $\rho = 0.445$; partial correlation beyond $|\Delta\phi_0|$ positive at every N).

and hierarchical-population [19] formulations make explicit. The three-variable system integrated here is exact on the uniform-constants (Poisson) submanifold, where the collective dynamics closes independently of N ; the off-manifold constants, which a finite- N ensemble need not realize, are precisely where the residual $3.2\times$ prolongation must originate — sharpening, not softening, the open problem.

7 Chimera collapse on the nonlocal ring

7.1 Why the ring, and where on it

The mean-field system showed aging without lifetime growth; is either property an artifact of the all-to-all topology? The nonlocally coupled ring of Eq. (2) is the system in which finite-size chimera transience was first quantified: in the deep-stable regime its chimeras are chaotic transients whose lifetimes grow rapidly with oscillator count [4]. That regime is unmeasurably long-lived at feasible compute, so we measure the near-boundary window instead — $\beta = 0.110\text{--}0.130$, just inside the chimera existence boundary $\beta_c \approx 0.13\text{--}0.14$ — where every run dies within the horizon and survival analysis has full power. The campaign design, event criterion, primary endpoint, and decision rule were pre-registered before any production run (Sec. 2.2).

7.2 Ring collapse is a structured, aging hazard

Collapse on the ring is not a memoryless escape. Across all 15 pre-registered (β, N) cells the censored-Weibull shape $\hat{k}$ ranges from 1.13 to 1.68 with every 95% profile-likelihood interval excluding one; the likelihood-ratio test against the exponential rejects memorylessness in 15/15 cells (p from 2×10^{-2} down to 5×10^{-27}); the kernel-smoothed hazard rises visibly in every cell (max/min rise factors 1.46–2.27); and a runs test rejects linearity of $\ln \hat{S}(t)$ in 15/15 cells ($p \approx 10^{-27}\text{--}10^{-58}$). The shape is threshold-robust: re-detecting deaths on the stored traces over $\varepsilon_{\text{std}} \in \{0.03, 0.04, 0.05, 0.06\}$ moves $\hat{k}$ by $\lesssim 0.07$ with the interval excluding one at every threshold. Overall censoring is 0.8% (34/4,500; worst cell 9.3% at $\beta = 0.110$, $N = 32$; exactly zero at $N \geq 64$).

The shape strengthens both toward criticality and with size: at $N = 32$, $\hat{k}$ rises from 1.13 to 1.33 across the β sweep, and at $\beta = 0.130$ from 1.33 to 1.68 across $N = 32\text{--}128$. Median lifetime meanwhile *decreases* with N — at $\beta = 0.130$ from 872 to 345 time units over $N = 32\text{--}256$ (Fig. 10a). This decreasing ring median lifetime with N at these near-boundary β reflects the near-collapse side of the existence boundary rather than the deep-stable regime of Wolfrum–Omel’chenko [4], in which lifetimes grow with N ; the two regimes sit on opposite sides of the boundary and there is no contradiction between them.

7.3 The dying object is a canonical chimera

The lifetime inversion invites the worry that the ρ_{std} criterion times a non-canonical decay channel, so we validated the dying object directly. The spatial field $\rho_k(t, \cdot)$ was not stored in the campaign, but every run is exactly reproducible from its logged seed: we re-ran all 300 trajectories per cell at $N = 256$ for $\beta = 0.110$ and $\beta = 0.130$ with field dumping, and the re-detected lifetimes reproduce the campaign values bit-for-bit (0/600 mismatches). Classifying each pre-death state by its spatial structure — an equilibration gate on the contrast ($\max \rho_{\text{std}} > 0.08$ over the living window, i.e. a chimera ever established) together with the dominant spatial Fourier mode of ρ_k , which is drift-invariant — shows the death criterion times the collapse of a genuine chimera, one contiguous coherent arc plus one incoherent arc or a multi-headed variant, in 98.0% ($\beta = 0.110$) and 97.3% ($\beta = 0.130$) of runs (Fig. 11).

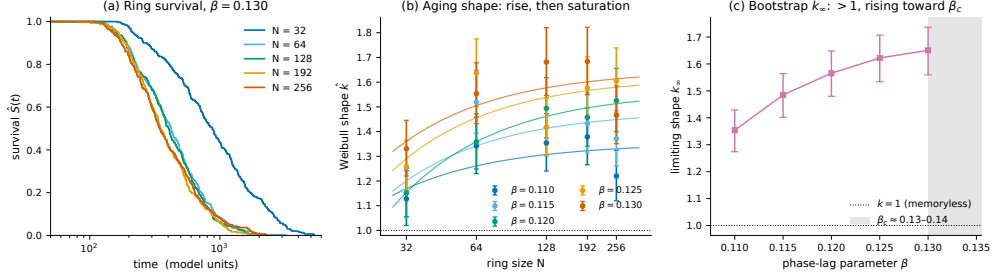


Fig. 10 Ring collapse ages, and its limiting shape is finite. (a) Kaplan–Meier survival by N at $\beta = 0.130$: median lifetime decreases with N ($872 \rightarrow 345$ over $N = 32 \rightarrow 256$) — the near-boundary side of the existence boundary, opposite to the deep-stable lifetime growth of Ref. [4]. (b) Censored-Weibull shape $\hat{k}(N)$ per β (profile-likelihood 95% intervals): all 25 (β, N) cells sit above the memoryless $k = 1$, rising with N and then saturating; curves are the AICc-selected bounded fits $k_\infty - a/N$ (Appendix C). (c) Bootstrapped k_∞ versus β : every interval excludes one, and the limiting shape rises toward the existence boundary $\beta_c \approx 0.13\text{--}0.14$.

The degenerate “dissolve-then-linger” channel — the chimera disperses early and a low-contrast state persists until full homogenization — is real but rare, 2.0–2.7%. Refitting with it removed leaves the Weibull shape unchanged or slightly larger ($\hat{k} = 1.22 [1.12, 1.32] \rightarrow 1.24 [1.14, 1.35]$ at $\beta = 0.110$; $1.47 [1.35, 1.58] \rightarrow 1.48 [1.36, 1.60]$ at $\beta = 0.130$), with intervals still excluding one; sweeping the contrast gate over $[0.06, 0.12]$ only moves $\hat{k}$ further from one. The increasing-hazard claim is therefore not an artifact of the rare degenerate channel.

7.4 The hazard shape is criterion-independent

Nor is the result a property of the particular detector. Re-detecting death on the same $\beta = 0.130$ trajectories under alternative literature-style criteria — structure loss at $\rho_{\text{std}} < 0.08$ and < 0.10 , and a mean-coherence-level rule — the median lifetime decreases with N under every one ($\bar{\tau}(256)/\bar{\tau}(32) = 0.34\text{--}0.40$), so the inversion is a property of the near-boundary dynamics, not of the detector. The increasing-hazard shape $\hat{k}(N) > 1$ is likewise robust under the structure-loss criteria, with all intervals excluding one and 0% censoring at every N . A mean-coherence criterion (ρ -mean above a fixed level) appears to drive $\hat{k}$ below one at large N , but this is a diagnosed censoring artifact: 6.7% of the $N = 256$ collapses go to a spatially uniform *twisted* state ($\rho_k \approx 0.50$, genuinely dead by spatial homogeneity at finite time) rather than the high-coherence sync plateau, and a coherence-level threshold mislabels these genuine deaths as censored, injecting a spurious never-dying tail; re-timing them at their true deaths restores $\hat{k}(N) = 1.26, 1.44, 1.62, 1.54, 1.57$, all intervals above one (Appendix C, Fig. C3). An independent Kaplan–Meier log-cumulative-hazard slope (1.5–2.1 across N) corroborates the increasing hazard with no Weibull assumption at all. That the spatial-homogeneity criterion — unlike a coherence-level threshold — captures both the synchronized and the twisted collapse channels is itself the reason to prefer it.

CP2 (C2): what the $\rho_{\text{std}} < 0.04$ criterion kills at $N=256$ — 97–98% genuine chimera deaths, 2–3% degenerate channel
 filtered Weibull k unchanged (0.110: 1.220 → 1.241; 0.130: 1.466 → 1.481, CIs exclude 1)

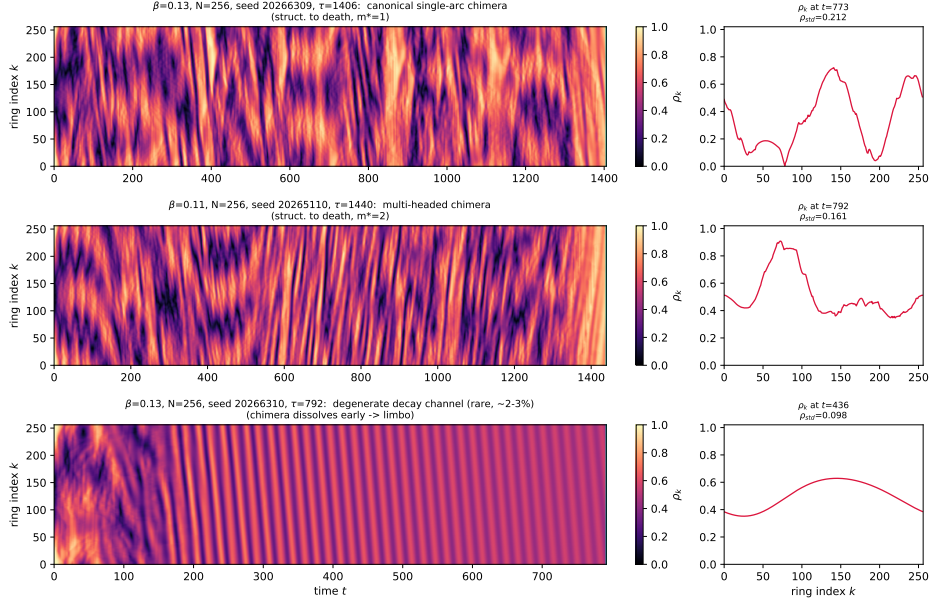


Fig. 11 What the $\rho_{\text{std}} < 0.04$ criterion kills at $N = 256$. Space–time local-coherence fields $\rho_k(t, k)$ (left) with a representative mature snapshot (right) for the three pre-death behaviors: a canonical single-arc chimera (top, structured to death), a multi-headed chimera (middle), and the rare degenerate channel (bottom, the chimera disperses early and a low-contrast state lingers). 97–98% of deaths are of the first two (genuine chimera) types; removing the degenerate channel leaves the Weibull shape unchanged.

7.5 Aging in the waiting time, not the descent

The structure lives in the waiting time, not in the final collapse. The terminal committed descent through the structure band is stereotyped — mean 52.8 time units, median 50.5, coefficient of variation 0.72 — and uncorrelated with lifetime (Spearman $|\rho| \leq 0.14$ in every cell) over lifetimes spanning 10^2 – 10^4 : the longer a ring chimera has lived, the more collapse-prone it has become, while the collapse itself, once committed, takes a characteristic time independent of the age at which it occurs. This is the ring’s version of the mean-field picture of Sec. 5: the hazard rises along the trajectory, and death itself is a fast, stereotyped event.

7.6 The limiting shape is finite and exceeds one

The rise of $\hat{k}$ with N poses the thermodynamic-limit question: does the shape saturate or diverge? Extending the campaign to $N = 192$ and 256 (300 runs per cell at all five β , 0% censoring) and selecting between bounded and divergent forms of $k(N)$ by AICc, the saturating form $k(N) = k_\infty - a/N$ wins in all five β cells (gaps 2.29–5.42), with bootstrapped $k_\infty = 1.35, 1.49, 1.56, 1.62, 1.65$ for $\beta = 0.110 \rightarrow 0.130$ —

every interval excluding one, rising monotonically toward β_c (Fig. 10b,c; Appendix C). The increasing-hazard character of ring chimera collapse is not a finite- N artifact: it survives the limit, with a limiting shape that sharpens toward criticality.

Together with Secs. 4–5, the comparison separates what is universal from what is topological. In both systems collapse ages — a hazard rising along the trajectory, expressed as a breath-locked ratchet in the mean field and as $k > 1$ at every size on the ring. The finite-size scaling, by contrast, is opposite: flat $\tau_{\text{abs}}(N)$ in the mean field, where the collective flow is N -free and N enters only through the initial-condition ensemble, versus decreasing median lifetime on the near-boundary ring (and growth in the deep-stable regime [4]), where the topology makes lifetime genuinely size-dependent. The aging belongs to the chimera; the scaling belongs to the network.

8 Discussion

The inversion, operationally. The deployed voice’s celebrated reliability — $\approx 96\%$ in-state over half-hour renders — is revealed as 100% supervisor-maintained: at the operating corner there is nothing to be reliable, since every unsupervised trajectory dies in ≈ 139 s regardless of how many oscillators are added. Conversely the corner discarded as unreliable holds the system’s only genuine attractor. For applications, two consequences follow. First, supervision is *intrinsic* to post-homoclinic operation at any N — re-seeding is not a finite-size patch but the entire persistence mechanism. Second, collapse detection should be absorption-grade: a detector with a recovery window (rather than the 2-s sustain rule, which over-triggers on 69–97% of firings) would intervene an order of magnitude less often for the same uptime.

The open problem: noise that prolongs. The reduced flow explains the geometry of death — where the corners sit, the breath period, the spiral rate, even individual lifetimes’ ordering — but captures $3.2\times$ too fast, and the flat $\tau_{\text{abs}}(N)$ plateau has no analog in an N -free collective flow. Both point the same direction: finite- N fluctuations *extend* the transient’s life relative to mean-field, the opposite of the usual intuition in which noise destroys metastable states. At the operating corner this prolongation is N -independent — the same factor of 3.2 across the full $16\times$ range of N — though that independence is itself a property of the corner’s depth beyond the homoclinic, not of every post-homoclinic parameter: at $\beta = 0.10$ it survives at the depth-matched point (coefficient of variation 0.14 across N) but weakens at the deeper $A = 0.5$, where the factor falls from 2.0 to 1.1 (Appendix A). A natural candidate for the prolongation is fluctuation-induced re-injection — finite-size kicks scattering the trajectory back toward the ghost region the deterministic spiral is leaving. We tested the simplest version directly (Appendix B): additive Gaussian noise of amplitude $c/\sqrt{N}$ on the collective variables of the reduced flow. It does produce re-injection and prolongation, but the prolongation is N -dependent — strong at small N and absent by $N = 64$ — because the noise amplitude is. The observed N -independent factor of 3.2 at the operating corner therefore rules out additive collective noise and points to a mechanism whose effective re-injection strength does not scale away as $1/\sqrt{N}$ (for example structured or multiplicative fluctuations tied to the breath cycle). We report the puzzle

as measured and leave the responsible mechanism open; the rising-hazard (aging-in-cycles) structure and the breath-locked phase distribution remain sharp constraints any such theory must satisfy.

Topology: aging transfers, scaling does not. On nonlocally coupled rings in the deep-stable regime, finite-size chimeras are chaotic transients with lifetimes growing rapidly in N [4]. The two-population mean-field case is qualitatively different: its collective flow is N -free, the N -dependence entering only through the initial-condition ensemble — and correspondingly we find no lifetime growth at all. Section 7 turns this contrast from a citation into a measurement. On the near-boundary side of the ring’s existence boundary, median lifetime *decreases* with N , under every death criterion tested, while the hazard keeps the same aging signature as the mean field: $\hat{k} > 1$ in every cell, rising with N and saturating to $k_\infty \approx 1.35\text{--}1.65 > 1$ (bootstrap intervals excluding one) in the thermodynamic limit. In both systems the aging is trajectory-level — surviving conditioning on the collective initial condition in the mean field (Sec. 4), and robust to the death criterion and to the identity of the dying object on the ring. The N -scaling of chimera transience is a property of the coupling topology; the aging of the collapse hazard is not.

Scope. All results are at strictly identical frequencies, equal population sizes, the shipped phase lag ($\beta = 0.05$, with the phenomenology reproduced at $\beta = 0.10$; Appendix A), and one canonical seed family; the absolute τ_{abs} carries the documented 22–27% labeling sensitivity; and the collective-coordinate summaries are a lossy projection of the $2N$ -phase state (weakest at $N = 4$, where they predict little). Within that scope, every headline claim is criterion-robust, bit-reproducible, and regenerable from committed configurations.

9 Conclusion

An engineering question — why does a synthesizer’s chimera need so much re-seeding? — unwound into a dynamical one. The collapse detector turns out to detect grazes, not deaths: most sustained near-synchrony excursions heal, and true absorption takes twice as long as first passage suggests. Corrected lifetimes are N -independent and aging, deaths lock to the breathing phase and arrive by a monotone ratchet of the envelope, and the bifurcation structure of the reduced two-population model explains why: the operating corner lies beyond the homoclinic, so the deployed state is a transient on a destroyed cycle’s ghost, while the discarded corner holds the true chimera attractor. A three-variable, parameter-free flow predicts the period, the spiral rate, and the ordering of individual deaths — and underestimates their timing by $3.2\times$, leaving one clean open problem: at finite N , fluctuations make this chimera live longer, not die faster.

Carrying the same survival methodology to the canonical ring substrate then shows which half of the story is universal. Near its existence boundary the ring’s chimera also dies by an aging process — Weibull shape above one in every pre-registered cell, saturating to $k_\infty \approx 1.35\text{--}1.65 > 1$ in the thermodynamic limit, with the dying object validated as a genuine chimera and the shape robust to the death criterion — while its lifetimes *fall* with N , opposite to the mean-field plateau and to the deep-stable

ring growth. Chimera collapse ages regardless of topology; how long the chimera lives, and how that lifetime scales with size, belongs to the network.

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Declarations

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Competing interests

The author declares no competing interests.

Data and code availability

The data and code that support the findings of this study — the integrators, experiment harnesses, campaign records, committed configurations, and figure-generation scripts — are openly available in the annealMusic repository at github.com/akarlin3/annealMusic and archived at Zenodo (doi.org/10.5281/zenodo.20564867); `python3 tools/paper-figures/run_all.py` regenerates the mean-field figures and `tools/paper-figures/fig10_ring.py` the ring summary figure. The ring campaign (pre-registration, ensemble records, and fits) is in `anneal-hazard/`; the stratified-hazard and criterion-robustness analyses, including the field re-runs validating the dying ring object, are in `critical_review/` and were independently reproduced with a disjoint censored-Weibull implementation.

Author contributions

Avery Karlin: Conceptualization (lead); Methodology (lead); Software (lead); Formal analysis (lead); Investigation (lead); Data curation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead).

Appendix A Robustness of the phenomenology at $\beta = 0.10$

To show the operating-corner phenomenology is not an artifact of the shipped phase lag $\beta = 0.05$, we repeated the finite- N campaign at $\beta = 0.10$, the value at which the

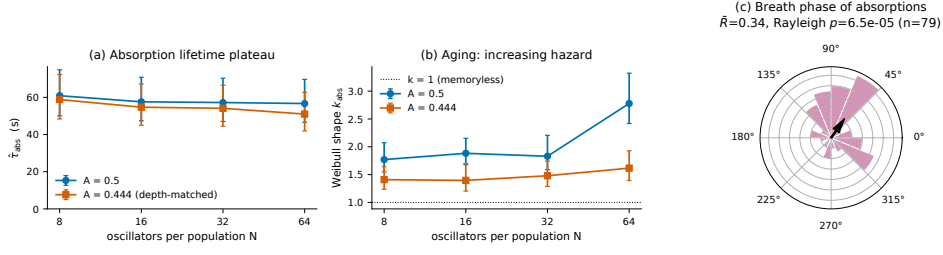


Fig. A1 Robustness at $\beta = 0.10$. (a) Absorption lifetime $\hat{\tau}_{\text{abs}}(N)$ is flat in N at both the post-homoclinic corner $A = 0.5$ and the depth-matched $A = 0.444$. (b) Censored-Weibull shape $k_{\text{abs}}(N)$ stays above the memoryless value $k = 1$ (aging persists). (c) True-absorption breath phases remain clustered (pooled $\bar{R} = 0.34$, Rayleigh $p = 6.5 \times 10^{-5}$, $n = 79$).

reduced model is validated against the published regimes [3]. Reduced-model placement gives a saddle-node at $A_{SN} = 0.178$, a Hopf at $A_H = 0.278$, and a homoclinic at $A_{hc}(0.10) = 0.354$ (bisection bracket width 9×10^{-5}). We sampled the post-homoclinic corner $A = 0.5$ (parallel to the operating corner) and, to match its distance beyond the homoclinic, a depth-matched point $A = A_{hc} + 0.090 = 0.444$; $A = 0.2$ and a cleaner mid-band point $A = 0.24$ sit in the stable-chimera band as never-absorber controls.

Every headline result survives (Fig. A1). The absorption lifetime is flat in N ($\hat{\tau}_{\text{abs}}(64)/\hat{\tau}_{\text{abs}}(8) = 0.93$ at $A = 0.5$; 0.87 at $A = 0.444$; fitted exponential rate $\approx -1 \times 10^{-3}$ per oscillator, zero censoring). The censored-Weibull shape stays above one ($k_{\text{abs}} = 1.8\text{--}2.8$ at $A = 0.5$, $1.4\text{--}1.6$ at $A = 0.444$), so hazard still rises. The per-cycle ratchet is positive ($\langle \Delta M \rangle = +0.06$ to $+0.19$, bootstrap CIs excluding zero), and true absorptions remain breath-phase-locked (pooled Rayleigh $p = 6.5 \times 10^{-5}$). The reduced flow reproduces the breath period ($\approx 23\text{--}25$ s, measured $20\text{--}25$ s) and spiral rate. The stable-band controls behave as expected: the mid-band $A = 0.24$ holds a near- N -independent persistent fraction (44% at $N = 8$, 52% at $N = 64$; 50 seeds each), a cleaner never-absorber control than $A = 0.2$, whose persistent fraction swings more widely (28% to 60%) as it sits close to A_{SN} .

One caveat refines a body-text claim. The transient is shallower at $\beta = 0.10$ (≈ 2 breath cycles to absorption vs ≈ 6 at the operating corner), and correspondingly the reduced-to-measured capture ratio — the $3.2\times$ prolongation of Sec. 8 — is N -independent only at comparable post-homoclinic depth. At the depth-matched $A = 0.444$ it stays flat (coefficient of variation 0.14 across N), but at the deeper $A = 0.5$ it falls from 2.0 to 1.1 across N . The N -independence of the prolongation is thus a property of the operating corner's depth beyond the homoclinic, not of every post-homoclinic parameter.

Appendix B A finite-size noise test of fluctuation re-injection

The reduced flow captures $\approx 3.2\times$ faster than the finite- N system, and at the operating corner that prolongation is N -independent (Sec. 8). The natural candidate —

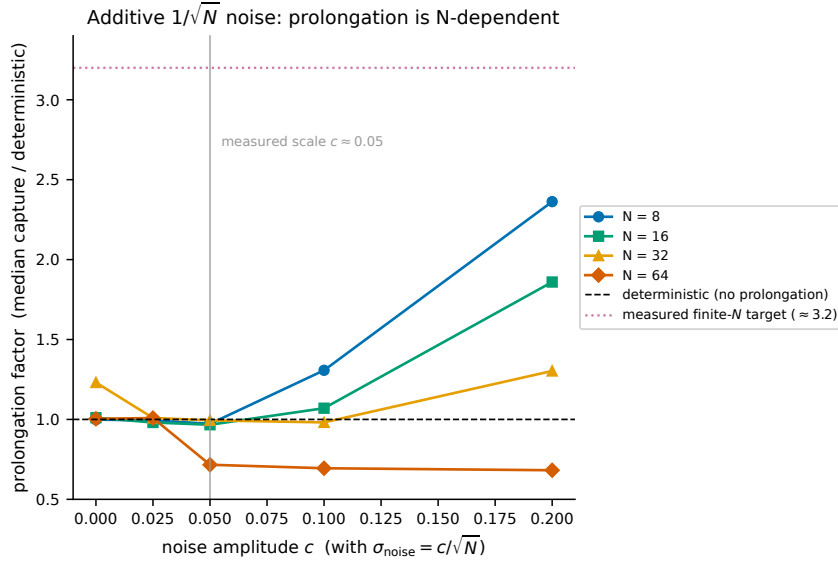


Fig. B2 Finite-size noise test. Prolongation factor (median capture relative to the deterministic reduced flow) versus additive-noise amplitude c ($\sigma_{\text{noise}} = c/\sqrt{N}$) at $A = 0.5$, $\beta = 0.05$. At the physical scale $c \approx 0.05$ the curves sit near unity; at larger c they fan out with N , so the prolongation is N -dependent and never reaches the measured ≈ 3.2 (dotted) where it is N -independent.

fluctuation-induced re-injection — predicts that adding finite-size noise to the collective flow should reproduce both facts. We tested this directly, integrating the three-variable reduced flow at the operating corner ($A = 0.5$, $\beta = 0.05$) with the Euler–Maruyama scheme and adding Gaussian noise of amplitude $\sigma_{\text{noise}} = c/\sqrt{N}$ to the collective variables (ρ_1, ρ_2, ψ) , with reflection keeping ρ in $[0, 1]$. At $c = 0$ the scheme reproduces the deterministic capture times to a median 0.25% per initial condition. The physical scale, estimated from the measured high-frequency fluctuation of R_{incoh} in the baseline data, is $c \approx 0.05$. We swept $N \in \{8, 16, 32, 64\}$, $c \in \{0, 0.025, 0.05, 0.1, 0.2\}$, 200 realizations each, from the Sec. 6 seed-mapped initial conditions.

The result is a clean negative (Fig. B2). At the physical scale $c \approx 0.05$ the noise barely perturbs capture (prolongation factor ≈ 0.97 , N -independent). Larger amplitudes do prolong — at $c = 0.2$ the trajectory is repeatedly re-injected, accumulating up to ~ 240 breath cycles before capture — but the prolongation is strongly N -dependent (factor 2.4 at $N = 8$ falling to 0.7 at $N = 64$; coefficient of variation 0.40 across N), and never reaches the measured ≈ 3.2 at any cell where it is N -independent. The rising-hazard constraint holds only at small c ($k > 1$ for $c \leq 0.05$, $k \approx 1$ for $c \geq 0.1$), while breath-phase locking survives throughout. Thus additive $1/\sqrt{N}$ noise on the collective flow reproduces the mechanism (re-injection) but with the wrong N -scaling: because its amplitude is N -dependent, so is the prolongation it produces. The N -independent $3.2\times$ factor requires a different mechanism.

Table C1 Saturate-versus-diverge model selection for the ring’s $\hat{k}(N)$, per β . k_∞ is the bounded-fit asymptote with bootstrap 95% interval; the gap is AICc(best divergent) – AICc(best saturating), decisive at ≥ 2 ; γ is the exponent of the confirmatory three-parameter saturating fit.

β	$\hat{k}(N = 32 \dots 256)$	k_∞ [95% CI]	AICc gap	γ
0.110	1.13, 1.34, 1.35, 1.38, 1.22	1.35 [1.27, 1.43]	+2.98	1.63
0.115	1.15, 1.52, 1.42, 1.43, 1.37	1.49 [1.40, 1.56]	+5.42	1.51
0.120	1.15, 1.36, 1.49, 1.46, 1.53	1.56 [1.48, 1.65]	+2.29	1.14
0.125	1.26, 1.64, 1.42, 1.58, 1.60	1.62 [1.53, 1.71]	+2.51	1.48
0.130	1.33, 1.55, 1.68, 1.68, 1.47	1.65 [1.56, 1.74]	+4.65	1.51

Appendix C Ring hazard-shape extrapolation

Section 7 reports that the ring’s Weibull shape $\hat{k}$ rises with N over the campaign sizes; this appendix settles whether that rise saturates or diverges as $N \rightarrow \infty$. The data are the per- (β, N) censored-Weibull shapes with profile-likelihood intervals over $N \in \{32, 64, 128, 192, 256\}$ (300 runs per cell). Censoring is exactly zero at $N \geq 64$ and at most 9.3% at $N = 32$; at the extrapolation anchors $N = 192$ and 256 the maximum observed lifetime is $\approx 4\text{--}6 \times 10^3$, far below $T_{\max} = 12,000$, so those fits are entirely uncensored.

Per β , we fit $k(N)$ by least squares weighted with the per-point uncertainty σ (the symmetrized profile interval), comparing one bounded form against two divergent ones: the saturating $k(N) = k_\infty - a/N$ (two parameters), the logarithmic $k(N) = b + a \ln N$, and the power law $k(N) = aN^\gamma$. Model selection is by AICc with the small-sample penalty at $n = 5$ points (two-parameter forms cost $\chi^2 + 10$, three-parameter forms $\chi^2 + 30$), and we call the comparison decisive when the AICc gap between the best divergent and best saturating form is at least 2. Uncertainty on k_∞ is bootstrapped: 5,000 Gaussian resamples of the $\hat{k}$ values within their σ , refit, percentile intervals.

The saturating form wins in all five β cells, with gaps 2.29–5.42 (Table C1). Adding $N = 192$ — the firming-up step that made a three-parameter fit AICc-valid — widened the three previously marginal gaps ($2.32 \rightarrow 2.98$ at $\beta = 0.110$; $2.10 \rightarrow 2.29$ at $\beta = 0.120$; $3.87 \rightarrow 4.65$ at $\beta = 0.130$) and weakened none. The general three-parameter form $k(N) = k_\infty - aN^{-\gamma}$ serves as confirmation rather than selection (its third parameter costs +20 AICc at $n = 5$): its k_∞ agrees with the bounded fit to within 0.02–0.04 everywhere, and its $\gamma \approx 1.1\text{--}1.6 > 1$ says the saturation is at least as fast as $1/N$, so the two-parameter form’s fixed $\gamma = 1$ is a slightly conservative choice. The nominally divergent fits barely diverge in any case: the fitted logarithmic slopes are 0.05–0.18 and the power exponents 0.04–0.13. All five bootstrap k_∞ intervals exclude one and are tight (width ≤ 0.18), and k_∞ rises monotonically toward the existence boundary: $1.35 < 1.49 < 1.56 < 1.62 < 1.65$ for $\beta = 0.110 \rightarrow 0.130$.

The Sec. 7 validations carry over to the extrapolation anchors. At $N = 256$ the dying object is a canonical chimera in 97–98% of runs (equilibration-gated classification of the re-run fields), and the equilibration-filtered refits — dropping runs in which no chimera ever established together with the rare degenerate channel — leave $\hat{k}$ unchanged or slightly larger, intervals still excluding one. The hazard shape is

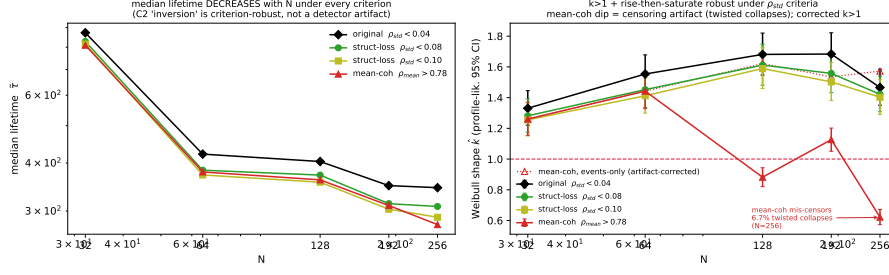


Fig. C3 Criterion-independence at $\beta = 0.130$. Left: median lifetime decreases with N under every death criterion (the near-boundary inversion is not a detector artifact). Right: the Weibull shape $\hat{k}(N)$ stays above one under the spatial-homogeneity (ρ_{std}) criteria; the mean-coherence rule's apparent dip below one at large N is a censoring artifact (twisted-state collapses mislabeled as censored), and the artifact-corrected curve returns above one.

criterion-robust across the spatial-homogeneity family ($\rho_{std} < 0.04, 0.08, 0.10$, all with 0% censoring), and the one criterion that appears to disagree — the mean-coherence level rule — is the diagnosed twisted-state censoring artifact of Sec. 7 (Fig. C3).

Two caveats bound the reading without weakening the call. First, four of the five cells are non-monotonic in $\hat{k}(N)$ (in-band scatter about the plateau; weighted $\chi^2 = 8$ –11 on 3 residual degrees of freedom), with only $\beta = 0.120$ a clean leveling fit ($\chi^2 = 0.70$): k_∞ should be read as the plateau *level*, ≈ 1.35 –1.68, not a precisely pinned asymptote. Second, the three-parameter fit is partially degenerate at five points (a and γ are not separately identified), which is why the headline rests on the parsimonious two-parameter form. What is firm is the binary outcome and its direction: the saturating form wins in 5/5 cells (0/5 divergent), and the limiting shape exceeds one in every cell — the increasing-hazard result survives the thermodynamic limit.

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